Predicting Bank Term Deposit Subscription Using Machine Learning

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***Abstract*—Predicting customer responses to bank marketing campaigns is important to improve customer targeting and reduce ineffective marketing efforts. This study investigates the application of machine learning algorithms to predict whether a customer will subscribe to a bank term deposit. Experiments were conducted using the UCI Bank Marketing dataset, which contains 45,211 customer records and 17 columns. To establish a realistic pre-call prediction setting, the call duration feature was excluded because it is only known after the current call. The resulting dataset contained 15 predictor features. Ten baseline classifiers, including a Dummy Classifier, were evaluated using a stratified 80:20 train-test split. Subsequently, hyperparameter tuning was performed on three selected models using stratified cross-validation. The tuned Random Forest model achieved the highest F1-score of 0.4626 and a ROC-AUC of 0.8018, with an accuracy of 84.99%. Extra Trees achieved the highest recall of 60.02%. Permutation feature importance identified month, contact method, housing loan status, previous campaign outcome, and contact day as the five most influential predictors. The findings demonstrate the potential of machine learning for identifying prospective term deposit subscribers, while also highlighting the challenges associated with imbalanced classification.**

***Index Terms*—Bank marketing, term deposit subscription, machine learning, classification, Random Forest, feature importance, class imbalance.**

I. Introduction

Banks use different marketing campaigns to promote financial products. One common product is a term deposit, where customers deposit money for a fixed period and receive interest. Banks often contact many customers to encourage them to subscribe to these products. However, not every customer is interested in the offer. As a result, marketing campaigns may require additional time, effort, and cost.

Machine learning can help address this problem by learning patterns from previous customer information. A trained model can use customer details and campaign history to predict whether a customer is likely to subscribe to a term deposit. This can help banks focus their marketing activities on potential customers.

Several researchers have studied bank marketing prediction using machine learning. Moro et al. introduced a data-driven approach to predict the success of bank telemarketing campaigns [1]. Later studies investigated different classification algorithms, ensemble learning methods, and prediction techniques for bank deposit subscription [2]–[7]. Other studies also explored data analysis, visualization, and machine learning-based customer targeting [8]–[15].

However, predicting customer subscription remains challenging because the Bank Marketing dataset contains significantly more customers who did not subscribe than customers who subscribed. Therefore, a model may achieve high accuracy while failing to identify many actual subscribers. For this reason, evaluation metrics such as precision, recall, F1-score, and ROC-AUC are important.

Another important issue is the availability of information at prediction time. The call duration feature is strongly related to the final subscription outcome, but it is only known after the call ends. If the goal is to predict customer interest before making a call, using this feature would create an unrealistic prediction setting. Therefore, this study excludes call duration from the input features, following the dataset documentation [16].

The main objectives of this study are:

1) To explore customer and campaign information in the Bank Marketing dataset.
2) To compare different machine learning classification algorithms.
3) To improve selected models through hyperparameter tuning.
4) To evaluate model performance using different classification metrics.
5) To identify important features and analyze prediction errors.

The main contribution of this study is a comparative evaluation of multiple classifiers under a pre-call prediction setting. It also combines model evaluation, statistical comparison, feature importance, and error analysis.

II. Related Work

Several studies have explored the use of machine learning to predict customer responses to bank telemarketing campaigns.

Moro et al. proposed a data-driven approach to predict the success of bank telemarketing campaigns using customer and campaign-related information [1]. Their work provided an important foundation for further research using the Bank Marketing dataset.

Later studies investigated different machine learning algorithms and ensemble learning techniques for predicting bank deposit subscriptions [2]–[7]. These studies explored how classification models can support customer targeting and improve prediction performance.

Other researchers examined visualization, cost-sensitive learning, decision trees, and machine learning frameworks for bank marketing prediction [8]–[13]. More recent studies have also continued to investigate machine learning approaches for predicting term deposit subscriptions [14], [15].

Although previous studies have achieved promising results, predicting customer subscription remains challenging because the dataset contains an imbalanced target variable. This study addresses the problem by comparing ten classification models, excluding call duration to maintain a pre-call prediction setting, and analyzing model performance using multiple evaluation metrics.

III. Methodology

A. Dataset Description

This study used the UCI Bank Marketing dataset [16]. The dataset contains 45,211 customer records and 17 columns, including customer information, campaign details, and the target variable. The target variable, `y`, indicates whether a customer subscribed to a term deposit.

The `duration` feature was excluded because it represents the duration of the current call and is not available before contacting a customer. After removing this feature and the target variable from the input data, 15 predictor features remained.

B. Data Preprocessing

The dataset was checked for missing values and duplicate records. No missing values or duplicate records were found. Numerical and categorical features were processed separately.

Numerical features were handled using an imputation and scaling pipeline, while categorical features were processed using imputation and one-hot encoding. All preprocessing steps were fitted using only the training data to prevent data leakage.

C. Train-Test Split

The dataset was divided into training and testing sets using an 80:20 ratio. Stratified sampling was applied to maintain the original class distribution in both sets. The training set contained 36,168 records, while the testing set contained 9,043 records.

D. Machine Learning Models

Ten classification models were evaluated in this study. These included Logistic Regression, Decision Tree, Random Forest, Extra Trees, Gradient Boosting, AdaBoost, K-Nearest Neighbors, Linear Support Vector Machine, Multi-Layer Perceptron, and Dummy Classifier.

The Dummy Classifier was included as a baseline to compare the performance of the machine learning models against a simple prediction strategy.

E. Hyperparameter Tuning

After the initial model comparison, Random Forest, Extra Trees, and Decision Tree were selected for hyperparameter tuning. RandomizedSearchCV was applied with stratified three-fold cross-validation. The tuning process used only the training data.

F. Evaluation Metrics

Model performance was evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and Precision-Recall AUC. These metrics were selected because the target variable was imbalanced. Confusion matrices were also used to examine correct and incorrect predictions.

IV. Results and Discussion

A. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the dataset and identify patterns in customer information and marketing campaigns. Five visualizations were used to examine the target distribution, numerical feature correlations, customer age distribution, job categories, and monthly subscription rates.

The target distribution showed that most customers did not subscribe to the term deposit. The correlation heatmap indicated that most numerical features had weak correlations. The age distribution showed that most customers were between 30 and 50 years old.

The job-wise and month-wise analyses also showed differences in subscription rates across customer groups and campaign months.



Fig. 1. Distribution of Target Classes

B. Initial Model Performance

Ten classification models were evaluated using the test dataset. Their performance was compared using accuracy, precision, recall, F1-score, and ROC-AUC.

The results showed that the Multi-Layer Perceptron achieved the highest accuracy of 89.51%. However, its recall

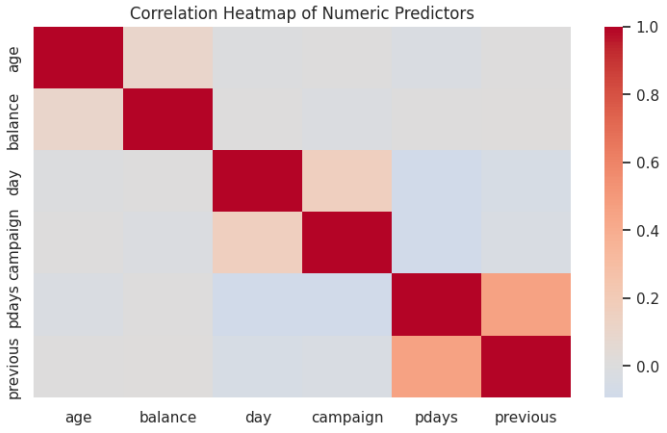


Fig. 2. Correlation Heatmap of Numerical Features

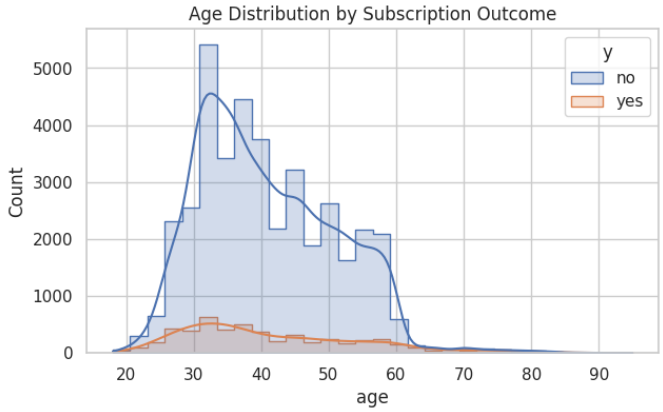


Fig. 3. Age Distribution of Customers

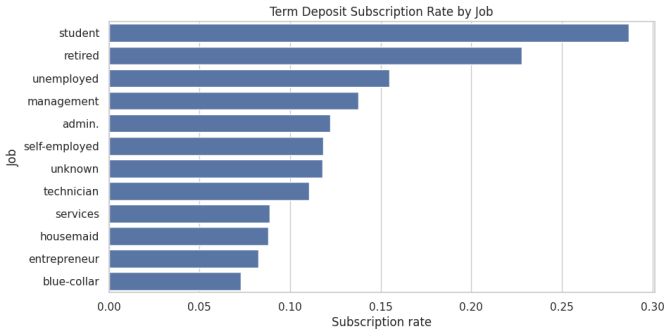


Fig. 4. Subscription Rate by Job Category

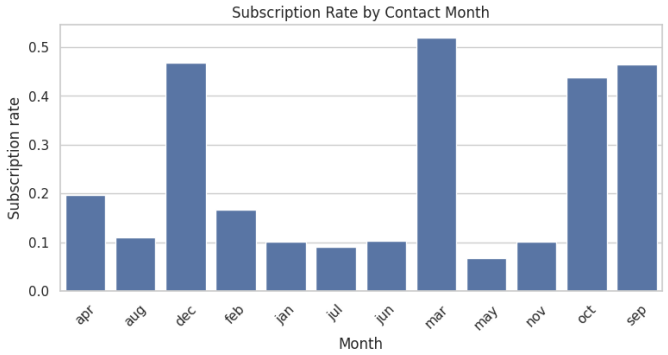


Fig. 5. Subscription Rate by Month

was only 25.99%, indicating that it failed to identify many actual subscribers.

Logistic Regression achieved the highest recall of 62.38% among the initial models. Extra Trees and Random Forest achieved the highest F1-score of 0.4494. These findings indicate that accuracy alone is not sufficient to evaluate model performance on this imbalanced dataset.

The Dummy Classifier achieved an accuracy of 88.30%, but its recall and F1-score were zero. This demonstrates why evaluating multiple performance metrics is important for this classification problem.

C. Hyperparameter Tuning Results

Hyperparameter tuning was performed on three selected models using RandomizedSearchCV with stratified three-fold cross-validation. The tuning process was conducted using only the training data.

Among the three tuned models, Random Forest achieved the highest F1-score of 0.4626. It also obtained the highest precision of 0.3981, ROC-AUC of 0.8018, and PR-AUC of 0.4530. Extra Trees achieved the highest recall of 60.02%, indicating its ability to identify more actual subscribers.

Based on the overall F1-score and other evaluation metrics, Random Forest was selected as the best overall model in this experiment.

D. ROC and Precision-Recall Curve Analysis

The ROC curve was used to evaluate the ability of the tuned Random Forest model to distinguish between subscribers and non-subscribers. The model achieved a ROC-AUC score of 0.8018.

The Precision-Recall curve was also examined because the target variable was imbalanced. The tuned Random Forest model achieved a PR-AUC score of 0.4530.

E. Confusion Matrix Analysis

Confusion matrices were used to examine the prediction performance of the three tuned models. The results show the number of correctly and incorrectly classified customers in the test dataset.

TABLE I
PERFORMANCE COMPARISON OF INITIAL MACHINE LEARNING MODELS

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC	PR-AUC
Extra Trees	0.8531	0.4003	0.5123	0.4494	0.7801	0.4181
Random Forest	0.8870	0.5226	0.3941	0.4494	0.8026	0.4548
Decision Tree	0.8316	0.3520	0.5227	0.4207	0.7196	0.3500
Linear SVM	0.7578	0.2684	0.6200	0.3746	0.7718	0.4049
Logistic Regression	0.7548	0.2662	0.6238	0.3732	0.7722	0.4093
MLP	0.8951	0.6236	0.2599	0.3669	0.7907	0.4430
KNN	0.8891	0.5758	0.1975	0.2942	0.7138	0.3434
Gradient Boosting	0.8942	0.6784	0.1815	0.2864	0.7940	0.4397
AdaBoost	0.8935	0.7039	0.1550	0.2541	0.7666	0.3961
Dummy Classifier	0.8830	0.0000	0.0000	0.0000	0.5000	0.1170

TABLE II
PERFORMANCE OF TUNED MODELS

Metric	RF	ET	DT
Accuracy	0.8499	0.8190	0.8223
Precision	0.3981	0.3434	0.3407
Recall	0.5520	0.6002	0.5548
F1-score	0.4626	0.4369	0.4222
ROC-AUC	0.8018	0.7906	0.7440
PR-AUC	0.4530	0.4293	0.3770

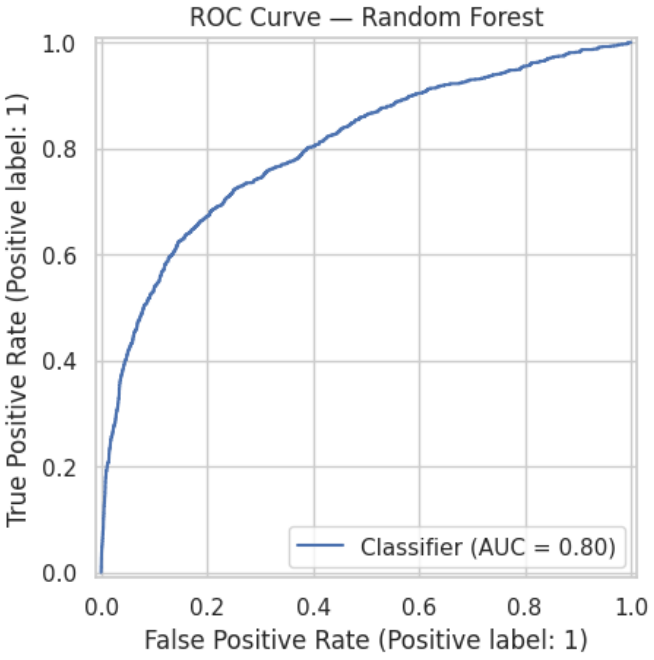


Fig. 6. ROC Curve of the Tuned Random Forest Model

For the Random Forest model, 7,102 non-subscribers and 584 subscribers were correctly classified. However, 883 non-subscribers were incorrectly predicted as subscribers, while 474 actual subscribers were incorrectly classified as non-subscribers.

Extra Trees correctly identified 635 actual subscribers, achieving the highest number of true positives among the three models. It also produced the lowest number of false negatives

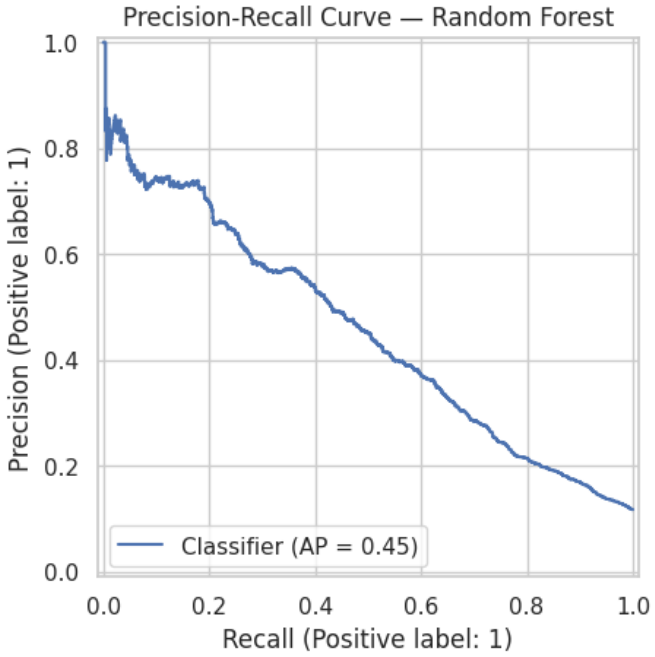


Fig. 7. Precision-Recall Curve of the Tuned Random Forest Model

(423). These results indicate that Extra Trees was better at identifying actual subscribers, while Random Forest achieved the highest overall F1-score.

TABLE III
CONFUSION MATRIX RESULTS OF TUNED MODELS

Model	TN	FP	FN	TP
Random Forest	7102	883	474	584
Extra Trees	6771	1214	423	635
Decision Tree	6849	1136	471	587

F. Feature Importance Analysis

Permutation feature importance was used to identify the most influential features in the tuned Random Forest model. The importance was measured by observing the change in F1-score after randomly shuffling each feature.

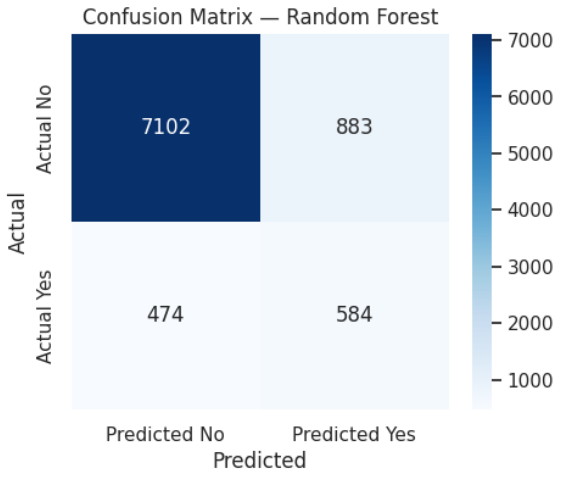


Fig. 8. Confusion Matrix of the Tuned Random Forest Model

The results showed that month was the most influential feature, followed by contact method and housing loan status. Previous campaign outcome and contact day were also among the five most important features.

These findings suggest that campaign timing, contact method, and previous campaign information played important roles in predicting customer subscription. However, feature importance indicates predictive contribution and does not establish a causal relationship.

TABLE IV
TOP FIVE FEATURES BASED ON PERMUTATION IMPORTANCE

Feature	Importance	Std. Dev.
Month	0.0999	0.0036
Contact	0.0930	0.0015
Housing	0.0379	0.0051
Poutcome	0.0289	0.0030
Day	0.0225	0.0067

G. Statistical Comparison

McNemar's exact test was conducted to compare the prediction outcomes of the Random Forest and Extra Trees models. The test produced a p-value of 8.94×10^{-35}, which is lower than the significance level of 0.05. This indicates a statistically significant difference between the prediction outcomes of the two models.

Extra Trees correctly classified 410 cases where Random Forest made incorrect predictions. In contrast, Random Forest correctly classified 130 cases where Extra Trees made incorrect predictions. These results show that the two models made significantly different prediction errors on the test dataset.

H. Error Analysis

Error analysis was performed to examine the prediction mistakes made by the tuned Random Forest model. Out of 9,043 test instances, the model correctly classified 7,686 cases. It produced 883 false positives and 474 false negatives.

False positives represent customers who did not subscribe but were incorrectly predicted as potential subscribers. False negatives represent actual subscribers who were incorrectly classified as non-subscribers.

The higher number of false positives indicates that the model incorrectly identified some non-subscribing customers as potential subscribers. These errors may lead to ineffective marketing efforts. On the other hand, false negatives represent missed potential customers. Therefore, reducing both types of errors is important for improving the practical usefulness of the model.

V. CONCLUSION

This study investigated the use of machine learning models to predict bank term deposit subscriptions. The Bank Marketing dataset was used to train and evaluate ten machine learning models. The results showed that the tuned Random Forest model achieved the highest F1-score of 0.4626, while Extra Trees achieved the highest recall of 0.6002 among the tuned models.

The study also identified month, contact method, housing loan status, previous campaign outcome, and contact day as the five most influential features. The findings indicate that machine learning can help identify potential customers for bank marketing campaigns. However, the class imbalance and prediction errors remain important challenges.

In future work, advanced sampling techniques, class-weighted learning, and additional optimization methods can be explored to improve the prediction of potential subscribers.

REFERENCES

[1] S. Moro, P. Cortez, and P. Rita, "A Data-Driven Approach to Predict the Success of Bank Telemarketing," Decision Support Systems, vol. 62, pp. 22–31, 2014.

[2] S. Hou, Z. Cai, J. Wu, H. Du, and P. Xie, "Applying Machine Learning to the Development of Prediction Models for Bank Deposit Subscription," International Journal of Business Analytics, vol. 9, no. 1, 2022. doi: 10.4018/IJBAN.288514.

[3] C. Xie, J.-L. Zhang, Y. Zhu, B. Xiong, and G.-J. Wang, "How to Improve the Success of Bank Telemarketing? Prediction and Interpretability Analysis Based on Machine Learning," Computers & Industrial Engineering, 2023. doi: 10.1016/j.cie.2022.108874.

[4] Y. Feng, Y. Yin, D. Wang, and L. Dhamotharan, "A Dynamic Ensemble Selection Method for Bank Telemarketing Sales Prediction," Journal of Business Research, vol. 139, pp. 368–382, 2022. doi: 10.1016/j.jbusres.2021.09.067.

[5] M. Peter, H. Mofi, S. Likoko, J. Sabas, R. Mbura, and N. Mduma, "Predicting Customer Subscription in Bank Telemarketing Campaigns Using Ensemble Learning Models," Machine Learning with Applications, vol. 19, Art. no. 100618, 2025. doi: 10.1016/j.mlwa.2025.100618.

[6] S. Moro, P. Cortez, and P. Rita, "Using Customer Lifetime Value and Neural Networks to Improve the Prediction of Bank Deposit Subscription in Telemarketing Campaigns," Neural Computing and Applications, vol. 26, no. 1, pp. 131–139, 2015. doi: 10.1007/S00521-014-1703-0.

[7] J.-M. Yu and S.-B. Cho, "Prediction of Bank Telemarketing with Co-training of Mixture-of-Experts and MLP," in Proc. 23rd International Conference on Neural Information Processing (ICONIP), 2016, pp. 52–59.

[8] A. Abu-Srhan, B. Alhammad, S. Al Zghoul, and R. Al-Sayyed, "Visualization and Analysis in Bank Direct Marketing Prediction," International Journal of Advanced Computer Science and Applications, vol. 10, no. 7, pp. 651–657, 2019. doi: 10.14569/IJACSA.2019.0100785.

[9] N. Ghatasheh, H. Faris, I. AlTaharwa, Y. Harb, and A. Harb, "Business Analytics in Telemarketing: Cost-Sensitive Analysis of Bank Campaigns Using Artificial Neural Networks," Applied Sciences, vol. 10, no. 7, Art. no. 2581, 2020. doi: 10.3390/app10072581.

[10] F. Safarkhani and S. Moro, "Improving the Accuracy of Predicting Bank Depositor's Behavior Using a Decision Tree," Applied Sciences, vol. 11, no. 19, Art. no. 9016, 2021. doi: 10.3390/app11199016.

[11] S. C. K. Tékouabou, S. C. Gherghina, H. Toulni, P. Neves Mata, M. N. Mata, and J. M. Martins, "A Machine Learning Framework towards Bank Telemarketing Prediction," Journal of Risk and Financial Management, vol. 15, no. 6, Art. no. 269, 2022. doi: 10.3390/jrfm15060269.

[12] W. Guo, Y. Yao, L. Liu, and T. Shen, "A Novel Ensemble Approach for Estimating the Competency of Bank Telemarketing," Scientific Reports, vol. 13, Art. no. 20819, 2023. doi: 10.1038/s41598-023-47177-7.

[13] F. Nasir, A. A. Ahmed, M. S. Kiraz, I. Yevseyeva, and M. Saif, "Data-Driven Decision-Making for Bank Target Marketing Using Supervised Learning Classifiers on Imbalanced Big Data," Computers, Materials & Continua, vol. 81, no. 1, pp. 1703–1728, 2024. doi: 10.32604/cmc.2024.055192.

[14] S. Du, "Findings from the Bank Marketing Dataset: Using Machine Learning to Forecast Term Deposit Subscriptions," Advances in Economics, Management and Political Sciences, vol. 240, pp. 62–69, 2025. doi: 10.54254/2754-1169/2025.BL29270.

[15] X. Wu, "Machine Learning Methods for Bank Term Deposit Subscription Prediction," International Journal of Global Economics and Management, vol. 10, no. 4, pp. 36–43, 2026. doi: 10.62051/ijgem.v10n4.06.

[16] S. Moro, P. Rita, and P. Cortez, "Bank Marketing," UCI Machine Learning Repository, 2014. doi: 10.24432/C5K306.